SKYDET: An End-to-End Multiscale Attentive Detection Network From Foundation Models for Small Objects in Remote Sensing Images

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***Abstract*—Remote sensing object detection remains a highly challenging task due to drastic variations in target scale, dense distributions of small objects, and complex background interference. Although existing detectors have achieved notable progress, they heavily rely on large-scale supervised pretraining and often suffer from substantial domain gaps and expensive annotation costs. In recent years, vision foundation models (VFMs) have demonstrated strong general-purpose representation capability, yet their potential in remote sensing imagery has not been fully explored. To bridge this gap, we propose SKYDET, an end-to-end robust object detection framework that explicitly migrates the billion-parameter DINOv3 model to the aerial domain. To effectively bridge the domain gap and prevent representation manifold degradation, we freeze the pretrained foundation model and propose a semantic guiding adapter (SGA) that acts as a precise semantic filter to suppress irrelevant background clutter. In addition, to address feature misalignment and semantic ambiguity during cross-scale fusion, we introduce a cross-fused encoder (CFE), whose core component is the reciprocal guidance module (RGM). The RGM establishes a reciprocal enhancement mechanism that enables spatial structure and channel semantics to guide each other, thereby effectively suppressing background noise and strengthening responses to small objects. Extensive experiments on three challenging benchmark datasets—DOTA-v1.0, AI-TOD, and NWPU VHR-10—demonstrate highly competitive performance. Specifically, SKYDET-C achieves state-of-the-art (SOTA) detection precision (with AP_{50} scores of 72.6%, 56.1%, and 95.6%, respectively), while SKYDET-T exhibits superior advantages in high-precision localization tasks. This validates the effectiveness of transferring VFM knowledge to remote sensing tasks. Our code is available at https://github.com/zhangyyy-ai-rs/SKYDET**

***Index Terms*—Detection Transformer (DETR), remote sensing, small-target detection, vision foundation models (VFMs).**

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I. Introduction

With the rapid advancement of science and technology and continuous economic growth, the accelerating urbanization process has imposed higher requirements on fine-grained urban management, making the task of constructing smart cities using various types of geospatial information increasingly critical. As one of the most important sources of geospatial information, remote sensing imagery plays an irreplaceable role in a wide range of applications, including land surveys [1], [2], [3], military reconnaissance [4], [5], [6], disaster emergency response [7], [8], [9], and smart city development [10], [11], [12], owing to its unique observation perspective and relatively long imaging distance. With the continuous upgrading and iteration of sensors, massive amounts of remote sensing imagery can now be acquired for analysis. However, such large-scale data also bring significant interpretation challenges: how to automatically, accurately, and efficiently identify and localize geographic targets of interest (e.g., aircraft, ships, vehicles, and buildings) from these vast imagery collections remains a key challenge that urgently needs to be addressed.

Remote sensing object detection differs from natural-image detection and faces unique challenges, including significant scale variations, a predominance of small objects, and complex backgrounds. With the rapid development of deep learning, researchers have begun applying deep learning methods to remote sensing object detection to address these challenges. Convolutional neural network (CNN)-based detectors, represented by Faster R-CNN [13] and YOLO [14], employ end-to-end learning frameworks to enable efficient image recognition. However, the inherent inductive bias of CNNs limits their ability to model the long-range dependencies and global contextual information that are prevalent in

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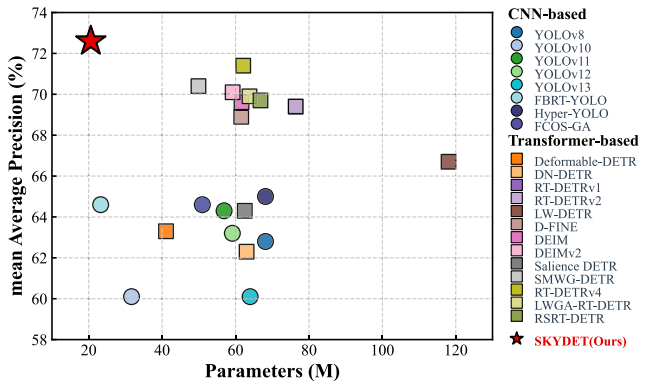


Fig. 1. Accuracy of representative detectors—comparison of computational parameters on the DOTA-v1.0 dataset—shows that our proposed SKYDET achieves a better balance between detection accuracy and computational cost, making it more suitable for target detection tasks in remote sensing images.

remote sensing imagery, thereby constraining the understanding of complex scenes and small objects. Recently, detection Transformer (DETR)-style models based on the Transformer architecture have been introduced to remote sensing object detection, offering a new direction to overcome these limitations. In particular, RT-DETR [15] is the first to focus self-attention on deep, semantically rich feature maps and combines it with convolutional operations to achieve cross-scale feature interaction, effectively enhancing the capture of global context. It substantially improves recognition accuracy while maintaining detection efficiency, providing a new solution for remote sensing object detection. Nevertheless, despite their different designs, the aforementioned CNN- and Transformer-based models share a common dependency on a powerful backbone network that is extensively pretrained on large annotated datasets. This dependency leads to two key issues: 1) model performance is constrained by domain discrepancies between the pretraining data and the target domain, and 2) acquiring large-scale, high-quality annotated data is costly. Meanwhile, existing models struggle to achieve a favorable balance between recognition speed and accuracy, which severely limits further progress [16], [17], [18], [19].

To overcome this bottleneck, the emergence of vision foundation models (VFMs) offers a new paradigm for computer vision. The general visual knowledge embedded in VFMs provides unprecedented opportunities to tackle the challenges of remote sensing object detection: their strong representation capability helps extract more discriminative features from complex backgrounds, and their superior generalization ability is expected to alleviate domain shift in remote sensing scenarios. Among them, the DINO [20], [21], [22] series has demonstrated that features learned from unlabeled natural images can transfer effectively to specialized domains. Moreover, the recently proposed DINOv3 addresses dense feature degradation in large-scale training and produces higher-quality feature representations through a unique training objective, outperforming existing models on tasks that require precise spatial understanding. This offers a highly promising technical route for leveraging its powerful feature representations to address the various challenges in remote sensing object detection.

However, how to effectively transfer the general representation capability of VFMs to remote sensing object detection and fully exploit their potential remains a critical yet underexplored problem. In this work, we conduct the first systematic evaluation of multiple DINOv3 variants in terms of their practical detection performance on remote sensing object detection. Building upon its strong feature extraction capability, we propose SKYDET, a novel DETR variant that, to the best of our knowledge, is the first to introduce DINOv3 into remote sensing object detection and incorporates multiple dedicated modules to maximize its detection performance in this domain. To validate the effectiveness of SKYDET, we conduct experiments on three challenging remote sensing datasets. Compared with existing mainstream methods, our model establishes a robust and highly competitive baseline for foundation model adaptation, providing a new paradigm for remote sensing object detection (Fig. 1).

Our main contributions are as follows.

1) We systematically evaluate, for the first time, the feature extraction capability of different components of DINOv3 in remote sensing object detection, and further propose, to the best of our knowledge, the first end-to-end Transformer-based detection model—SKYDET—that incorporates DINOv3 for remote sensing object detection. By effectively leveraging its strong feature representations, SKYDET significantly improves the detection accuracy and robustness for remote sensing targets.
2) To fully exploit the key feature information extracted by the ViT in DINOv3, we design an adapter, termed semantic guiding adapter (SGA), which effectively converts the single-scale features of DINOv3 into multiscale representations and additionally supplements fine-grained information in the imagery to enrich semantic content, thereby providing more informative feature representations for the detection stage.
3) To maximize the feature extraction capability of DINOv3 for remote sensing object detection, we propose a cross-fused encoder (CFE), which effectively integrates key target features with low computational overhead, thereby improving the robustness of the model.

II. Related Work

A. Vision Foundation Models

VFMs refer to visual models pretrained on extremely large-scale datasets, featuring strong general-purpose visual representation capability and in-context learning ability [23]. Essentially, they can be viewed as a more generalized form of self-supervised learning, and the learned feature space exhibits favorable semantic alignment properties, offering new possibilities for addressing intraclass variation and interclass similarity in complex scenes. VFMs have mainly evolved along two directions. The first comprises general-purpose visual models represented by the DINO series, which obtain highly robust and transferable visual representations via self-supervised learning on hundreds of millions of images, and

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are particularly effective for pixel-level dense prediction tasks. DINOv2 [21] demonstrates that pure self-supervised learning on a large curated dataset can yield high-quality features suitable for dense prediction. Building on this, DINOv3 [22] further scales the model parameters to 7 billion while expanding the data scale to 1.7 billion, effectively alleviating dense feature degradation and marking a new milestone in general visual representation learning. The second direction is vision–language models represented by CLIP [24], which align visual and language modalities through contrastive learning and exhibit strong zero-shot inference and semantic understanding capability. By contrastive learning on hundreds of millions of web image–text pairs, CLIP maps images and texts into a unified semantic space, and its strong zero-shot transfer ability lays the foundation for open-vocabulary recognition. Subsequent studies further refine CLIP's global image–text alignment to region- and pixel-level granularity. RegionCLIP [25] introduces region–text alignment pretraining, substantially extending CLIP to learn region-level visual representations and enabling open-vocabulary object detection for the first time. From another perspective, CRIS [26] adopts a vision–language decoder and a text-to-pixel contrastive loss, allowing the CLIP framework to learn pixel-level information and significantly improving performance on pixel-level segmentation conditioned on natural language descriptions. Despite the immense potential of VFMs, directly applying them to remote sensing object detection remains challenging. Fully fine-tuning these massive models on relatively small aerial datasets inevitably leads to catastrophic forgetting and representation manifold degradation. Unlike existing methods, our SKYDET proposes a novel paradigm by freezing the DINOv3 backbone to preserve its general-purpose representations, while employing specially designed adapters to bridge the domain gap.

B. Small-Target Detection in Remote Sensing Images

In remote sensing imagery, small objects are numerous yet occupy only a very small proportion of the entire image, which may cause their features to correspond to only a few pixels on the output feature maps; this is a key factor leading to missed detections and false alarms. Therefore, how to effectively exploit critical information from remote sensing images becomes a necessary condition for improving recognition accuracy in remote sensing detection tasks. Some studies focus on the backbone network by enhancing its feature extraction capability to provide the neck with more informative features for integration. Zhong et al. [25] proposed a novel local-to-global Transformer backbone, which adopts a hierarchical sparse attention mechanism and applies different attention schemes to shallow and deep features, thereby improving feature extraction and strengthening small-object representations without significantly increasing computational overhead. From another perspective, Lu et al. [27] leveraged the complementary advantages of convolution and pooling to better preserve subtle small-object cues during downsampling and further employed multibranch and feature decoupling strategies to improve multiscale feature extraction, providing a new solution for small-object detection in remote sensing imagery. Other works instead target the neck network, aiming to enhance cross-scale fusion to better aggregate multiscale information extracted by the backbone. Overlock [28] improves upon conventional pyramid networks by introducing a branched architecture with three collaborative subnetworks for parallel feature processing, effectively enhancing fine-grained fusion of dense small-object features. Hyper-YOLO [29] goes beyond traditional fusion paradigms by mapping visual features into a semantic space and constructing a hypergraph for information propagation, enabling complex high-order interactions across layers and spatial positions and achieving efficient integration of multiscale information. Very recently, specialized architectural designs have further advanced this field by addressing specific challenges in aerial imagery. For instance, to tackle the severe feature degradation of tiny targets, recent studies have proposed high-resolution feature generators specifically optimized for small-ship detection in optical images [30]. In the realm of label-efficient learning, novel multiinstance mining frameworks equipped with dynamic localization mechanisms have been developed to significantly improve weakly supervised object detection without relying on costly bounding-box annotations [31]. Furthermore, for objects with arbitrary orientations, dense representative points have been successfully utilized to explicitly guide the precise regression of rotated bounding boxes, demonstrating exceptional performance in dense ship detection [32].

However, a critical distinction remains between these methodologies and our proposed SKYDET. While the aforementioned methods propose exquisite modules for fine-grained feature extraction and precise localization, they predominantly rely on models trained from scratch or initialized with standard supervised weights. Consequently, their overall representation capacity is fundamentally bottlenecked by the limited scale and high intraclass variance of available remote sensing datasets. In stark contrast, SKYDET explicitly breaks this data-scale bottleneck by introducing the pretrained VFM into the remote sensing domain, avoiding feature degradation through a parameter-frozen strategy and bridging the representation gap via our proposed adapting modules.

C. DETR and Its Variants

As a pioneer of the Transformer-based end-to-end object detection paradigm, DETR discards the traditional anchor design and NMS postprocessing, and instead directly models input features via set prediction and bipartite matching, demonstrating strong feature extraction potential in complex backgrounds. Building upon this framework, researchers have developed numerous variants. Deformable DETR [33] introduces deformable attention to enable dynamic selection of feature locations and adopts multilevel feature fusion to reduce computational cost and accelerate convergence. DN-DETR [34] designs an auxiliary supervision task with coordinate perturbations, effectively improving convergence speed and detection robustness while bypassing reliance on bipartite matching. Based on this, the improved version DINO [35] further optimizes the denoising anchor-box mechanism and

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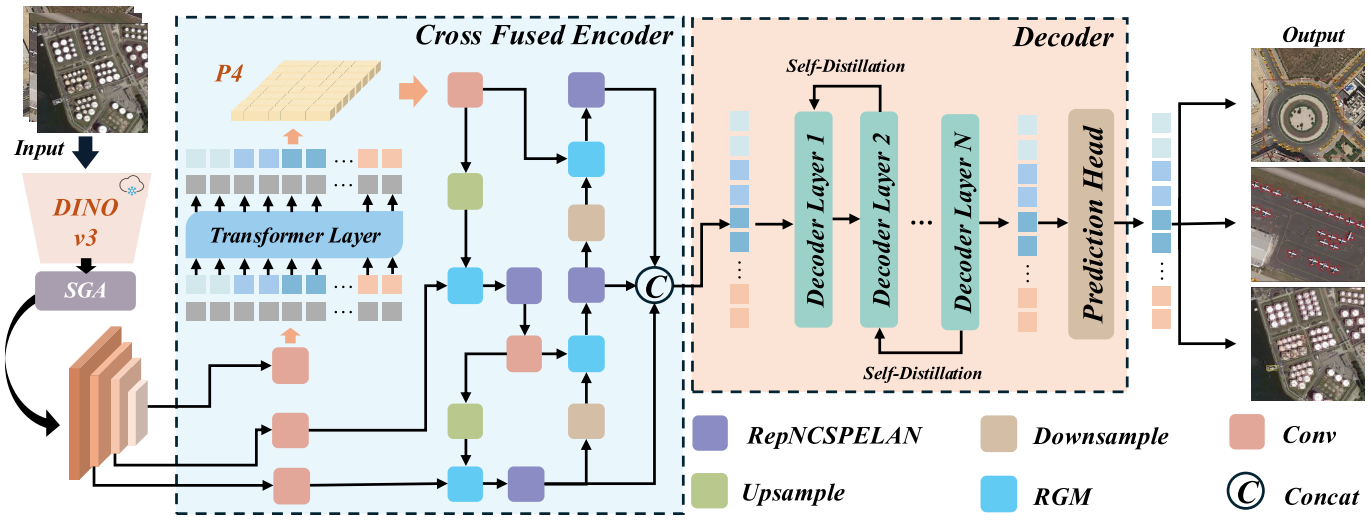


Fig. 2. Overall architecture of SKYDET comprises three key components: a backbone network, a cross-fused encoder, and a decoder with a self-distillation mechanism. First, the input image is fed into the backbone network composed of DINOv3, which leverages its powerful general feature representation capabilities to extract multilevel visual features. These features are then fed into the cross-fused encoder, where they are split into multiple paths for processing. After deep interaction and feature fusion, the features from each path are concatenated and sent to the decoder. In the decoder stage, the model uses multiple stacked decoding layers, introducing a self-distillation mechanism between layers to enhance the learning ability of feature representations. Finally, after processing by the prediction head, the model outputs refined bounding boxes for the remote sensing scene and confidence scores for each category, achieving accurate target detection.

has become a representative strong baseline. AO2-DETR [36] incorporates oriented box parameters into the Transformer architecture and addresses feature alignment for objects with large aspect ratios through rotation-aware attention. ARS-DETR [37] employs SkewIoU to dynamically adjust label smoothing, effectively alleviating the angle periodicity issue in oriented bounding box prediction. On the other hand, to meet the requirements of real-time performance and dense detection, RT-DETR [15] proposes an efficient hybrid encoder, substantially improving both detection speed and accuracy in Transformer-based detectors. D-FINE [38] redefines bounding-box regression and, through a fine-grained distribution refinement (FDR) mechanism, further advances the tradeoff between real-time efficiency and boundary precision. DEIM [39] focuses on the matching challenge in dense scenes; by combining a dense one-to-one matching strategy with a matching-aware loss, it not only accelerates convergence but also improves matching performance in crowded scenarios. While these DETR variants excel in natural imagery, they often struggle with feature misalignment and semantic ambiguity when directly applied to the complex backgrounds of remote sensing imagery. SKYDET builds upon the efficient paradigms of real-time DETRs but fundamentally distinguishes itself by introducing the SGA and CFE. These modules are specifically engineered to mitigate the domain gap by reciprocally fusing multiscale VFM features, establishing a robust and high-precision detector uniquely tailored for dense aerial scenes.

III. Methods

A. Overall Architecture of SKYDET

The architecture of SKYDET is illustrated in Fig. 2. As a DETR variant, it consists of three components: a feature extraction network, a CFE, and a decoder. First, the input image is fed into the feature extraction network, where the strong feature extraction capability of the pretrained DINOv3 is leveraged to perform stage-wise downsampling and generate multiscale feature representations. Inspired by RT-DETR, we design CFE to effectively integrate these features. Since the deepest features extracted by the backbone contain rich semantic information, we introduce SGA to act as a precise semantic filter, effectively suppressing background clutter and isolating targets before subsequent multiscale fusion. Moreover, to address spatial misalignment, the reciprocal guidance module (RGM) is embedded to enable fine-grained, reciprocal cross-scale interactions, successfully reconstructing the blurred boundaries of tiny objects. In the feature decoder stage, the fused multilevel features are encoded into dynamic object queries, and a Transformer is further used to generate detection boxes, thereby establishing a one-to-one correspondence between the predicted bounding boxes and the ground-truth labels.

B. DINOv3 Backbone

Traditional supervised backbones often face generalization bottlenecks in aerial scenarios due to limited labeled data and domain gaps. To address these limitations, we introduce DINOv3 as the backbone of SKYDET. Although DINOv3 is predominantly pretrained on generic natural images, its learned representations are inherently suited for mitigating the core challenges of remote sensing object detection: highly complex backgrounds and extremely small objects. Driven by the self-distillation paradigm, DINOv3 exhibits a powerful emergent property of object-centric localization. This acts as a robust semantic before separating foreground targets from chaotic aerial backgrounds. Furthermore, DINOv3 provides a resolution-agnostic and robust feature space that perfectly aligns with the structural features of tiny targets.

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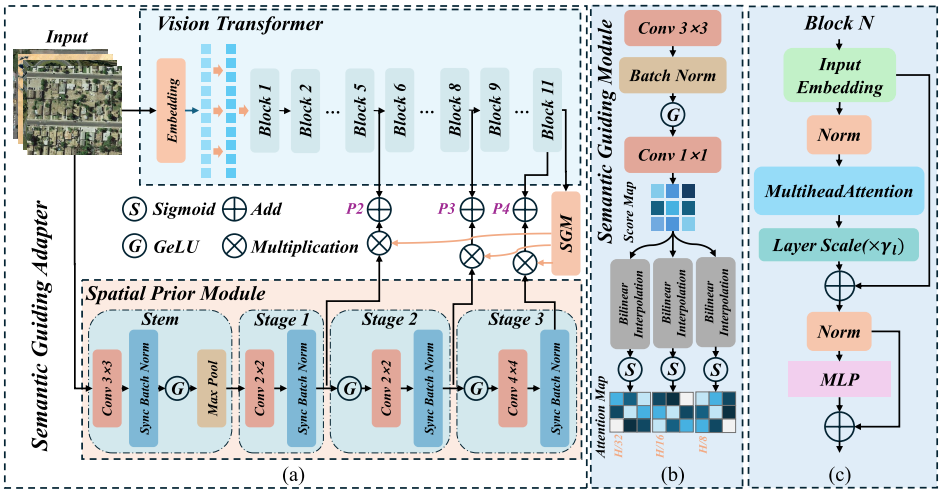


Fig. 3. (a) Overall architecture of SGA. (b) Main architecture of SGM and the core component of SGA. (c) Main architecture of a block in the ViT branch of SGA.

When integrating DINOv3 into SKYDET, we consider two design choices. The first is the network architecture: the DINOv3 framework supports two backbones, a Transformer-based ViT and a CNN-based ConvNeXt. The ViT benefits from self-attention for global context modeling, whereas ConvNeXt has inherent advantages in capturing local features. The second is the training strategy, that is, end-to-end fine-tuning versus fully freezing the backbone. The former updates all backbone parameters based on pretrained weights and has substantial potential for task-specific adaptation, but incurs additional computational overhead; the latter is an efficient alternative that keeps pretrained weights entirely frozen, allowing the model to exploit strong general representations learned from large-scale data without suffering from representation manifold degradation or catastrophic forgetting, thus greatly reducing training cost and preventing overfitting on limited aerial datasets. To select the optimal backbone for SKYDET, we conduct a comprehensive evaluation of these architectures and training strategies on remote sensing object detection. Notably, since the ViT produces only single-scale features, we additionally design an adapter for it (see Section IV-J). To address diverse application requirements, we introduce two complementary backbone configurations based on DINOv3, resulting in model variants with distinct design objectives: 1) SKYDET-C (utilizing the CNN-based ConvNeXt-Base), which leverages the inherent local inductive bias of CNNs to capture densely distributed, tiny objects, offering significant advantages in high-recall scenarios and practical deployments; and 2) SKYDET-T (utilizing the Transformer-based ViT), which employs self-attention mechanisms for long-range global context modeling to precisely refine boundaries, making it an ideal choice for scenarios demanding exceptional localization accuracy.

C. Semantic Guiding Adapter

As analyzed in Section III-B, although the ViT backbone exhibits strong feature extraction potential due to its powerful global context modeling capability, its inherent single-scale output directly conflicts with the hierarchical feature requirements of object detection. Therefore, an adapter is required to efficiently leverage the features extracted by the ViT. However, naive feature concatenation can introduce substantial background noise into ViT features, severely degrading the signal-to-noise ratio and consequently impairing recognition accuracy. To address this issue, we design SGA, which proactively uses high-level semantic knowledge from the ViT to guide and refine low-level spatial detail features before feature fusion. The overall architecture of SGA is shown in Fig. 3. It follows a dual-branch parallel design, consisting of a spatial branch that provides fine-grained low-level features and a semantic branch that provides global high-level understanding, jointly performing spatial decoupling and information extraction. Table I provides a detailed explanation of the subsequent parameters in this section. Given an input image $I \in \mathbb{R}^{3\times H\times W}$, the spatial branch employs a spatial prior module (SPM) to efficiently extract N-scale low-level spatial detail features $\{F^i_{\text{det}}\}_{i=1}^{N}$, which can be formulated as follows:

$$F^i_{\text{det}} = \text{SPM}(I), \quad i \in \{1, \ldots, N\} \tag{1}$$

TABLE I
MATHEMATICAL NOTATIONS AND TENSOR DIMENSIONS

Symbol	Description	Shape/Dim
I	Input remote sensing image	$R^{3\times H\times W}$
F^i_{det}	Spatial detail features from SPM at i-th scale	$R^{C_d\times H_i\times W_i}$
F^i_{sem}	Semantic features from ViT at i-th scale	$R^{C_s\times H_{sem}\times W_{sem}}$
M^i_{att}	Generated spatial attention map for i-th scale	$R^{1\times H_i\times W_i}$
$F^{i'}_{det}$	Refined spatial features after SGM guidance	$R^{C_d\times H_i\times W_i}$
F^i_{out}	Final fused multi-scale output features	$R^{C_{out}\times H_i\times W_i}$
σ	Sigmoid activation function	-
$\odot$	Element-wise (Hadamard) product	-
$[;]$	Concatenation along the channel dimension	-

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where F_{det} denotes the low-level spatial detail features. The semantic branch extracts N-scale high-level semantic features $\{F^i_{\text{sem}}\}_{i=1}^N$ from different intermediate layers of the ViT model. As illustrated in Fig. 3(c), the fundamental building block of the ViT architecture employs a layer scale mechanism. Specifically, layer scale introduces a set of learnable diagonal matrices to dynamically scale the residual features along the channel dimension. This operation effectively prevents feature collapse and significantly stabilizes the training process of the deep Transformer. Based on these features, to proactively suppress background noise and enhance target saliency before feature fusion, we design SGM that applies top-down attention weighting to the low-level spatial detail features. Fundamentally distinct from conventional FPN attention variants that rely on layer-by-layer local propagation (which is prone to noise accumulation), SGA establishes a single-source global guidance paradigm. Unlike methods that perform interactions independently at each scale, SGM uses the deepest semantic feature F^N_{sem} extracted by the ViT as the sole source of guidance. We argue that this feature, with the largest effective receptive field and the most abstract semantic understanding, provides the most informative cues for distinguishing large-scale target regions from background land-cover patterns. SGM distills it into a set of multiscale spatial attention maps $\{M^i_{\text{att}}\}_{i=1}^N$

$$M^i_{\text{att}} = \text{Upsample}\left(\sigma\left(\mathcal{G}_{\text{SGM}}\left(F^L_{\text{sem}}\right)\right)\right) \tag{2}$$

where $\mathcal{G}_{\text{SGM}}$ denotes the SGM network implemented with lightweight convolutional layers, σ is the Sigmoid activation function, and Upsample$(\cdot)$ upsamples the feature map to the same resolution (H_i, W_i) as the ith spatial feature. Each pixel value in the spatial attention map $M^i_{\text{att}} \in \mathbb{R}^{1\times H_i\times W_i}$ represents the semantic importance of the corresponding spatial location. We use it as a spatial attention gate and dynamically modulate the original spatial detail features via element-wise multiplication, yielding the enhanced features $F^{i\prime}_{\text{det}}$

$$F^{i\prime}_{\text{det}} = F^i_{\text{det}} \odot M^i_{\text{att}} \tag{3}$$

where $\odot$ denotes element-wise multiplication. Finally, the guided and enhanced spatial features are concatenated with the aligned semantic features and fed into the fusion layer $\mathcal{F}^i_{\text{fusion}}$ for deep fusion and channel projection, producing the multiscale output features F^i_{out} for the subsequent feature fusion network

$$F^i_{\text{out}} = \mathcal{F}^i_{\text{fso}}\left(\left[\text{Align}\left(F^i_{\text{sem}}\right); F^{i\prime}_{\text{det}}\right]\right) \tag{4}$$

where $[\cdot;\cdot]$ denotes concatenation along the channel dimension, and the Align$(\cdot)$ operator is used to adjust the resolution of semantic features to match that of the spatial branch. Through this design, SGA effectively transforms the single-scale features extracted by the ViT into representations with a higher signal-to-noise ratio and tighter integration of semantic and spatial information, providing a practical paradigm for SKYDET to fully exploit ViT's feature extraction capability.

D. Cross Fused Encoder

In real-time object detection, the DETR paradigm, by introducing hybrid encoders, has successfully decoupled multiscale feature processing and established a new performance benchmark. These encoders typically follow a design philosophy of intrascale feature interaction and cross-scale feature fusion. However, most mainstream methods still rely on simple concatenation to aggregate adjacent-level features during cross-scale fusion, which inevitably introduces semantic ambiguity and feature misalignment in the fused representations—an issue that is particularly pronounced in remote sensing detection with complex background noise. To address this problem, we propose CFE. As depicted in Fig. 2, for intrascale feature interaction, we construct the basic neck network utilizing the RepNCSPELAN module [40]. By integrating re-parameterization techniques with cross-stage partial (CSP) connections, it significantly enhances gradient propagation and feature reusability. For cross-scale feature fusion, we design RGM as the core component of the encoder. The latter can be formulated as a bidirectional process of feature calibration and reconstruction. As shown in Fig. 4, given an input reference feature $X_h \in \mathbb{R}^{C\times H\times W}$ (from deep layers) and a target feature $X_l \in \mathbb{R}^{C\times H\times W}$ (from shallow layers), RGM first constructs cross-domain attention descriptors to capture complementary information. Considering that $\mathbf{X}_h$ contains rich semantic localization priors while $\mathbf{X}_l$ preserves fine-grained texture details, we process these two features in different ways. For the spatial dimension, we extract a spatial saliency map $\mathbf{M}_s \in \mathbb{R}^{1\times H\times W}$ from $\mathbf{X}_h$ by aggregating channel information

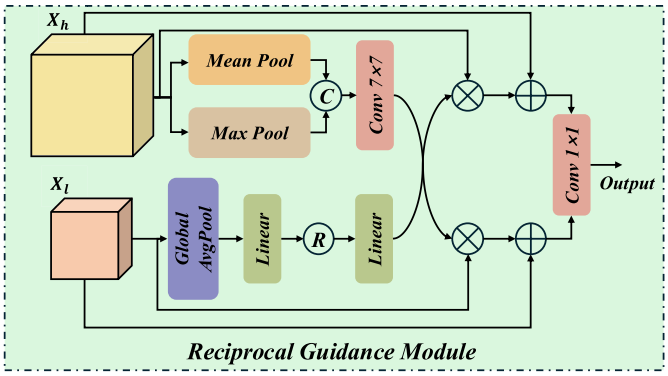


Fig. 4. RGM's modular architecture.

$$M_s = \sigma\left(f^{7\times7}\left(\left[\text{AvgPool}_s(X_h); \text{MaxPool}_s(X_h)\right]\right)\right) \tag{5}$$

where σ denotes the Sigmoid activation function, $f^{7\times7}$ denotes a convolution operation with a 7×7 kernel, $[\cdot;\cdot]$ indicates concatenation along the channel dimension, and AvgPool_s and MaxPool_s denote average pooling and max pooling along the channel dimension, respectively. Meanwhile, to capture key detail patterns in low-level features, we adopt a squeeze-and-excitation strategy to generate a channel attention vector $M_c \in \mathbb{R}^{C\times1\times1}$ from X_l

$$M_c = \sigma\left(W_2 \cdot \delta\left(W_1 \cdot \text{GAP}(X_l)\right)\right) \tag{6}$$

where $M_c \in \mathbb{R}^{C\times1\times1}$ denotes the generated channel attention vector, GAP denotes global average pooling, δ denotes the ReLU activation function, $W_1 \in \mathbb{R}^{C/r\times C}$ and $W_2 \in \mathbb{R}^{C\times C/r}$ are the weight matrices of a multilayer perceptron (MLP), and r is

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the reduction ratio. After obtaining the complementary attention maps, we design a cross-guidance mechanism. Instead of direct concatenation, RGM applies the spatial weights $\mathbf{M}_s$ to the shallow feature $\mathbf{X}_l$ and the channel weights $\mathbf{M}_c$ to the deep feature $\mathbf{X}_h$, and injects the original features via residual connections to prevent gradient vanishing. Let $\widetilde{X}_h$ and $\widetilde{X}_l$ denote the enhanced deep and shallow branches, respectively; this interaction can be formulated as follows:

$$\tilde{X}_h = X_h + (X_h \odot M_c) \tag{7}$$

$$\tilde{X}_l = X_l + (X_l \odot M_s) \tag{8}$$

where $\odot$ denotes element-wise multiplication with broadcasting. This design enables the $\widetilde{X}_h$ branch to absorb spatially filtered fine-grained details, while the $\widetilde{X}_l$ branch incorporates channel-selected semantic context. Finally, the two enhanced branches are concatenated and fused via a 1×1 convolution layer, yielding the final output $Y \in \mathbb{R}^{C\times H\times W}$

$$Y = f^{1\times 1}\left(\left[\widetilde{X}_h; \widetilde{X}_l\right]\right) \tag{9}$$

where $f^{1\times 1}$ denotes a 1×1 convolution operation, which maps the channel dimension from $2C$ back to C to keep the output dimension consistent with the input features. Through this process, RGM achieves adaptive alignment and complementary enhancement of cross-scale features. This interactive fusion strategy effectively alleviates the semantic ambiguity commonly observed in remote sensing imagery, providing the subsequent detection head with higher-quality feature representations that contain less background noise and exhibit clearer target boundaries.

IV. Experiments

A. Experimental Platform

All experiments in this work are conducted on Ubuntu 18.04, with model training and evaluation performed using four NVIDIA A10 PCIe GPUs. We use Python 3.10, PyTorch 2.1.0, and CUDA 12.0.

B. Datasets

To effectively evaluate the performance of SKYDET on remote sensing object detection, we conduct experiments on three representative remote sensing datasets: DOTA-v1.0 [41], NWPU VHR-10 [42], and AI-TOD [43].

1) DOTA-v1.0: DOTA is a popular dataset in remote sensing object detection, consisting of 2806 high-resolution optical remote sensing images and 188 282 object instances spanning 15 common categories. Following the long-side representation scheme, we crop the original images into subimages of 1024×1024 pixels with an overlap of 200 pixels, and remove subimages without any object annotations, resulting in 13 554 subimages in total. Meanwhile, we generate horizontal ground-truth boxes by minimizing the horizontal enclosing rectangles of the original ground-truth boxes. Finally, 10 276 images are used for training, and 3278 images are used for validation.

2) AI-TOD: The AI-TOD dataset consists of 28 036 aerial images acquired from diverse scenes and viewpoints, with image sizes ranging from 800×800 to 1400×1400 pixels, covering 700 621 object instances. We follow the official split provided by the dataset, using 14 536 images for training, 4272 images for validation, and 9228 images for testing.

3) NWPU VHR-10: As a public benchmark for object detection in high-resolution remote sensing imagery, NWPU VHR-10 contains 800 images with 3651 annotated object instances. The spatial resolution ranges from 0.5 to 2 m. We follow the official dataset split, using 550 images for training, 100 images for validation, and 150 images for testing.

C. Evaluation Metrics

To highlight the higher sensitivity of SKYDET to small objects, we adopt the COCO [59] absolute scale-based definition of object size at the beginning of our experiments and use it throughout all experiments in this article. Specifically, objects with a pixel area smaller than 32×32 are defined as small, those with a pixel area between 32×32 and 96×96 are defined as medium, and those with a pixel area larger than 96×96 are defined as large. In addition, we use average precision (AP) as the primary evaluation metric to quantify model performance across categories. This metric reflects the tradeoff between precision and recall and is measured as the area under the precision–recall curve. The computation is given as follows:

$$AP = \sum_{i}^{m} P_i = \int_0^1 P(R)\, dR \tag{10}$$

where m denotes the number of classes in the object detection task, and P and R represent precision and recall, respectively. Specifically, recall measures the proportion of correctly identified positive samples among all ground-truth positives, while precision measures the proportion of correctly identified instances among all predicted positives, which can be expressed as follows:

$$Recall = \frac{TP}{TP + FN} \tag{11}$$

$$Precision = \frac{TP}{TP + FP} \tag{12}$$

where true positives (TPs) denote the number of correctly detected bounding boxes, false positives (FPs) denote the number of incorrectly detected bounding boxes, and false negatives (FNs) denote the number of missed ground-truth boxes during detection. Notably, all results in this article are evaluated using three AP metrics: IoU, IoU, and AP@[0.5:0.95] IoU. On this basis, we further report AP_S, AP_M, and AP_L to evaluate the AP for small-, medium-, and large-scale objects, respectively. In addition, we adopt parameters (the number of model parameters) and giga floating-point operations (GFLOPs) to assess model size and computational efficiency.

D. Comparative Experimental Parameter Settings

To ensure the reproducibility and fairness of our experiments, we strictly define the hardware environment

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TABLE II

PERFORMANCE COMPARISON WITH SOTA METHODS ON THE DOTA V1.0 VAL SET. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Model	mAP (%)	AP per category (%)														
		PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC
YOLOv8 [44]	0.628	0.864	0.558	0.604	0.560	0.528	0.716	0.788	0.519	0.771	0.383	0.630	0.677	0.788	0.419	0.616
YOLOv10 [45]	0.601	0.856	0.387	0.548	0.565	0.462	0.702	0.745	0.537	0.771	0.362	0.634	0.643	0.823	0.449	0.530
YOLOv11 [46]	0.643	0.865	0.581	0.605	0.553	0.542	0.731	0.758	0.558	0.776	0.396	0.670	0.656	0.813	0.529	0.613
YOLOv12 [47]	0.632	0.865	0.567	0.604	0.554	0.534	0.729	0.752	0.524	0.779	0.396	0.644	0.647	0.830	0.441	0.608
YOLOv13 [48]	0.601	0.856	0.387	0.548	0.565	0.462	0.702	0.745	0.537	0.771	0.362	0.634	0.643	0.823	0.449	0.530
FBRT-YOLO [49]	0.646	0.893	0.556	0.660	0.579	0.574	0.756	0.772	0.514	0.790	0.437	0.612	0.683	0.810	0.406	0.644
Hyper-YOLO [29]	0.650	0.896	0.612	0.646	0.572	0.567	0.731	0.768	0.523	0.785	0.386	0.661	0.701	0.832	0.428	0.642
FCOS-GA [50]	0.646	0.876	0.610	0.598	0.565	0.580	0.720	0.775	0.564	0.780	0.471	0.550	0.680	0.759	0.441	0.521
Deformable-DETR [33]	0.633	0.802	0.628	0.641	0.552	0.621	0.698	0.712	0.495	0.735	0.455	0.601	0.635	0.845	0.467	0.598
DN-DETR [34]	0.623	0.851	0.665	0.585	0.608	0.555	0.715	0.725	0.525	0.762	0.442	0.545	0.625	0.823	0.405	0.615
RT-DETRv1 [15]	0.694	0.888	0.650	0.711	0.591	0.634	0.725	0.805	0.542	0.800	0.451	0.689	0.690	0.921	0.675	0.637
RT-DETRv2 [51]	0.694	0.885	0.672	0.708	0.595	0.627	0.740	0.798	0.548	0.803	0.459	0.659	0.687	0.921	0.680	0.626
LW-DETR [52]	0.667	0.855	0.665	0.672	0.582	0.655	0.735	0.755	0.525	0.775	0.472	0.625	0.675	0.890	0.485	0.636
D-FINE [38]	0.689	0.885	0.682	0.726	0.609	0.639	0.749	0.769	0.565	0.820	0.483	0.677	0.701	0.932	0.456	0.646
DEIM [39]	0.696	0.896	0.665	0.744	0.613	0.657	0.754	0.758	0.563	0.816	0.478	0.672	0.708	0.934	0.534	0.656
DEIMv2 [53]	0.701	0.895	0.697	0.709	0.612	0.688	0.772	0.788	0.547	0.815	0.500	0.665	0.710	0.934	0.512	0.667
Salience DETR [54]	0.643	0.861	0.632	0.614	0.586	0.599	0.677	0.705	0.534	0.748	0.417	0.596	0.564	0.902	0.533	0.677
SMWG-DETR [55]	0.704	0.890	0.682	0.744	0.604	0.664	0.775	0.799	0.563	0.815	0.493	0.678	0.704	0.932	0.532	0.685
RT-DETRv4 [56]	0.714	0.895	0.690	0.742	0.607	0.665	0.775	0.798	0.560	0.812	0.492	0.660	0.705	0.930	0.685	0.695
LWGA-RT-DETR [57]	0.699	0.889	0.680	0.735	0.599	0.657	0.768	0.793	0.556	0.810	0.488	0.655	0.699	0.928	0.525	0.703
RSRT-DETR [58]	0.697	0.890	0.681	0.735	0.598	0.663	0.772	0.796	0.555	0.812	0.492	0.656	0.700	0.929	0.525	0.678
SKYDET-T	0.708	0.885	0.685	0.782	0.598	0.668	0.795	0.802	0.578	0.825	0.501	0.642	0.705	0.925	0.521	0.695
SKYDET-C	**0.726**	**0.898**	**0.712**	**0.815**	0.609	**0.695**	**0.821**	**0.815**	**0.594**	**0.838**	**0.519**	0.665	**0.725**	**0.942**	0.535	**0.706**

and follow a standardized comparison protocol. In our experiments, we develop two variants: SKYDET-T, which adopts the ViT model in DINOv3, and SKYDET-C, which adopts DINOv3-ConvNeXt. Both variants share the same hyperparameters. Specifically, the total training batch size is set to 32, and the training schedule lasts for 120 epochs, including 108 epochs with advanced data augmentation and 6 rounds of standard training (notably, one round of standard training contains two epochs). During the advanced augmentation stage, the model employs four augmentation strategies: random photometric distortion, random zoom-out, random IoU cropping, and random multiscale input. During the standard training stage, these augmentations are disabled for fine-tuning. We set the input image size to 640 $\times$ 640, the embedding dimension to 384, and the feed-forward network dimension to 2048. We adopt AdamW for optimization, set the base learning rate to 0.00025 and the weight decay to 0.000125, and apply EMA with a decay rate of 0.9999 to improve training stability. Notably, the backbone parameters are kept frozen to leverage the strong general representations of DINOv3, while only the adapter and detection head are updated to ensure efficiency. For loss weighting, the fine-grained localization loss $\mathbf{L_{FGL}}$ is set to 0.15, and the global optimization self-distillation loss $\mathbf{L_{DDF}}$ is set to 1.5, together with $\mathbf{L_{VFL}}$, $\mathbf{L_{Box}}$, and $\mathbf{L_{GIoU}}$ weighted by 1, 5, and 2, respectively. In addition, the model uses a 6-layer decoder, and in the FDR module, the hyperparameters **a** and **c** of the weighting function are fixed to 0.5 and 0.125, respectively, with the number of distribution bins **N** set to 32. Furthermore, to address potential concerns regarding experimental fairness, we strictly follow the principle of "optimal configuration for upper bound." Given that CNN-based detectors (e.g., YOLO series) and Transformer-based detectors (e.g., DETR family) have fundamentally different convergence behaviors, forcing a uniform epoch count or augmentation strategy would be suboptimal. Therefore, we reproduce all baselines using their officially recommended optimal settings (e.g., 300 epochs with Mosaic/MixUp for YOLO-based methods, and dedicated multiscale training for DETR-based methods) to evaluate their maximum potential. For all compared methods, we exclusively select the largest available model variant (Extra Large/X-scale). Additionally, recognizing the trend of VFMs, we include the latest DINOv3-based detectors (e.g., DEIMv2) in our comparison to establish a rigorous benchmark within the VFM paradigm.

E. Comparison With SOTA Methods

To comprehensively evaluate the overall performance of SKYDET, we conduct rigorous quantitative comparisons with representative state-of-the-art (SOTA) object detection methods. The compared models include, but are not limited to, CNN-based detectors from the YOLO family and their variants, as well as Transformer-based end-to-end detectors represented by RT-DETR, D-FINE, DEIM, and the most recent advances, including RT-DETRv4 and RSRT-DETR. To ensure a fair comparison, all models are reproduced and evaluated under the same hardware environment using unified standardized scripts. During data preparation, we apply an identical preprocessing pipeline to all three benchmark datasets, including a standard tiling strategy and consistent evaluation scripts, to eliminate performance biases caused by data distribution differences. For training configurations, we follow the principle of "officially recommended settings first," ensuring that each baseline converges under its optimal hyperparameters. For YOLO-style methods, we adopt the classic Mosaic augmentation and a 300-epoch training schedule; for Transformer-based models, we strictly follow their

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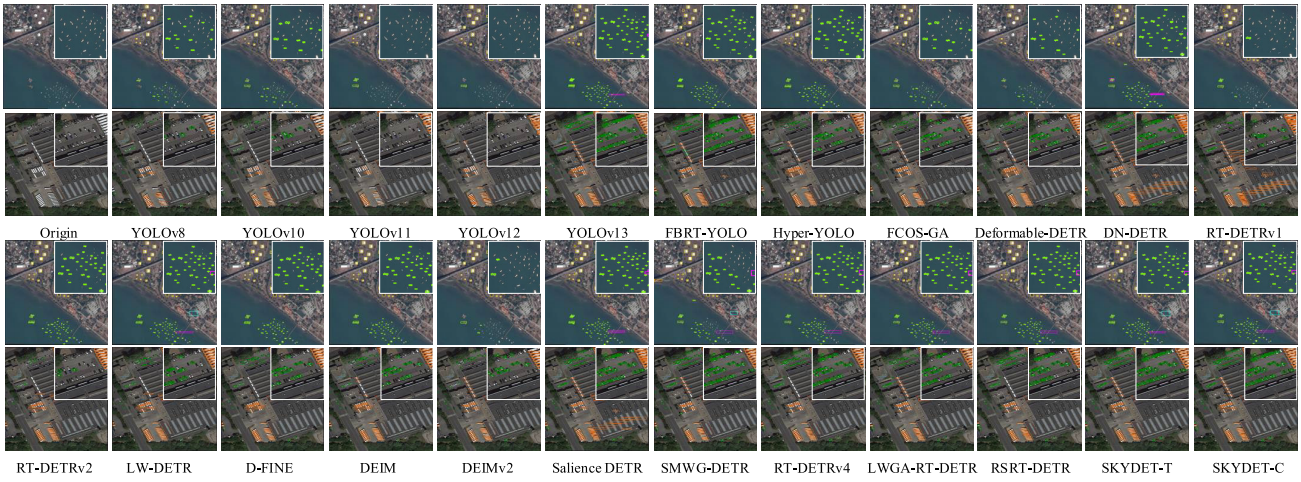


Fig. 5. Visualizing the detection results of SKYDET and other models on the DOTA dataset, it can be seen that SKYDET achieved the highest detection accuracy compared to other models.

TABLE III

PERFORMANCE COMPARISON WITH SOTA METHODS ON THE AI-TOD TEST SET. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Model	Params (M)	FLOPs (G)	AP_{50}	AP_{75}	$AP_{50:95}$
CNN Series					
YOLOv8 [44]	68.16	258.2	0.457	0.180	0.221
YOLOv10 [45]	31.67	171.1	0.452	0.173	0.212
YOLOv11 [46]	56.88	193.5	0.473	0.176	0.220
YOLOv12 [47]	59.13	199.9	0.423	0.143	0.188
YOLOv13 [48]	63.97	200.1	0.452	0.176	0.219
FBRT-YOLO [49]	23.26	187.1	0.421	0.158	0.198
Hyper-YOLO [29]	68.16	258.2	0.461	0.183	0.224
FCOS-GA [50]	50.95	234.3	0.478	0.185	0.228
Transformer Series					
Deformable-DETR [33]	41.06	175.4	0.385	0.105	0.124
DN-DETR [34]	63.02	283.5	0.402	0.118	0.135
RT-DETRv1 [15]	76.39	200.4	0.521	0.148	0.220
RT-DETRv2 [51]	76.38	200.3	0.523	0.152	0.152
LW-DETR [52]	118.03	174.2	0.373	0.098	0.152
D-FINE [38]	61.55	202.2	0.513	0.169	0.224
DEIM [39]	61.63	202.3	0.490	0.164	0.218
DEIMv2 [53]	59.22	182.4	0.535	0.178	0.231
Salience DETR [54]	62.52	205.8	0.457	0.125	0.191
SMWG-DETR [55]	65.9	194.1	0.552	**0.218**	**0.265**
RT-DETRv4 [56]	62.1	202.3	**0.553**	0.210	0.258
LWGA-RT-DETR [57]	63.8	184.6	0.536	0.188	0.238
RSRT-DETR [58]	66.8	203.6	0.542	0.201	0.248
SKYDET-T	805.5	458.2	0.547	**0.235**	**0.278**
SKYDET-C	20.58	237.8	**0.561**	0.212	0.264

TABLE IV

COMPARISON OF DETECTION ACCURACY ON THE NWPU VHR-10 TEST SET. BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED

Model	AP_S	AP_M	AP_L	AP_{50}	AP_{75}	$AP_{50:95}$
CNN Based						
YOLOv8 [44]	0.362	0.513	0.674	0.891	0.621	0.561
YOLOv10 [45]	0.367	0.486	0.671	0.865	0.598	0.535
YOLOv11 [46]	0.247	0.524	0.648	0.889	0.636	0.561
YOLOv12 [47]	0.335	0.512	0.683	0.901	0.601	0.560
YOLOv13 [48]	0.249	0.482	0.591	0.875	0.557	0.515
FBRT-YOLO [49]	0.270	0.489	0.587	0.822	0.574	0.526
Hyper-YOLO [29]	0.408	0.516	0.672	0.896	0.619	0.559
FCOS-GA [50]	0.385	0.522	0.685	0.905	0.615	0.568
Transformer Based						
Deformable-DETR [33]	0.285	0.531	0.612	0.852	0.584	0.510
DN-DETR [34]	0.301	0.545	0.628	0.871	0.605	0.528
RT-DETRv1 [15]	0.243	0.596	0.650	0.939	0.712	0.623
RT-DETRv2 [51]	0.368	0.577	0.658	0.929	0.697	0.617
LW-DETR [52]	0.346	0.545	0.677	0.922	0.676	0.597
D-FINE [38]	0.330	0.604	0.673	0.939	0.721	0.634
DEIM [39]	0.414	0.590	0.678	0.910	0.699	0.617
DEIMv2 [53]	0.385	0.569	0.654	0.901	0.703	0.606
Salience DETR [54]	0.425	0.605	**0.692**	0.903	0.728	0.635
SMWG-DETR [55]	0.470	0.630	0.685	**0.948**	0.760	0.670
RT-DETRv4 [56]	0.485	0.640	0.682	0.947	0.774	0.675
LWGA-RT-DETR [57]	0.438	0.616	0.665	0.940	0.742	0.651
RSRT-DETR [58]	0.520	0.635	0.675	0.943	0.790	0.682
SKYDET-T	**0.585**	**0.682**	**0.695**	0.945	**0.825**	**0.715**
SKYDET-C	**0.606**	**0.670**	0.651	**0.956**	**0.800**	**0.690**

dedicated multiscale training augmentations and prescribed training epochs. Notably, although different architectures vary in their sensitivity to input resolution, we ensure that each baseline is reproduced at its officially recommended optimal resolution, and we only compare the largest model size (extra large, *x*) for each method. Tables II–IV report detailed performance metrics of all models on DOTA-v1.0, NWPU VHR-10, and AI-TOD, respectively.

On the DOTA-v1.0 dataset, SKYDET demonstrates a clear performance advantage. As shown in Table II, SKYDET achieves 72.6% mAP, outperforming CNN-based detectors by 7.6%–12.5%, and surpassing Transformer-based detectors by 1.2%–10.3%. This improvement is not confined to a single category, but consistently observed across multiple scenarios. In particular, for categories that are prone to confusion with background textures, such as Bridge and Harbor, SKYDET effectively suppresses background noise through the reciprocal guidance between semantic features and spatial details in CFE, boosting the per-class AP to 81.5% and 94.2%, respectively. Moreover, benefiting from the strong visual representations of DINOv3, the model achieves high accuracy in detecting densely arranged objects such as small vehicles, while significantly reducing missed detections, validating the effectiveness of transferring VFMs to multiscale remote sensing tasks. Fig. 5 presents qualitative detection results of SKYDET and other methods on DOTA-v1.0.

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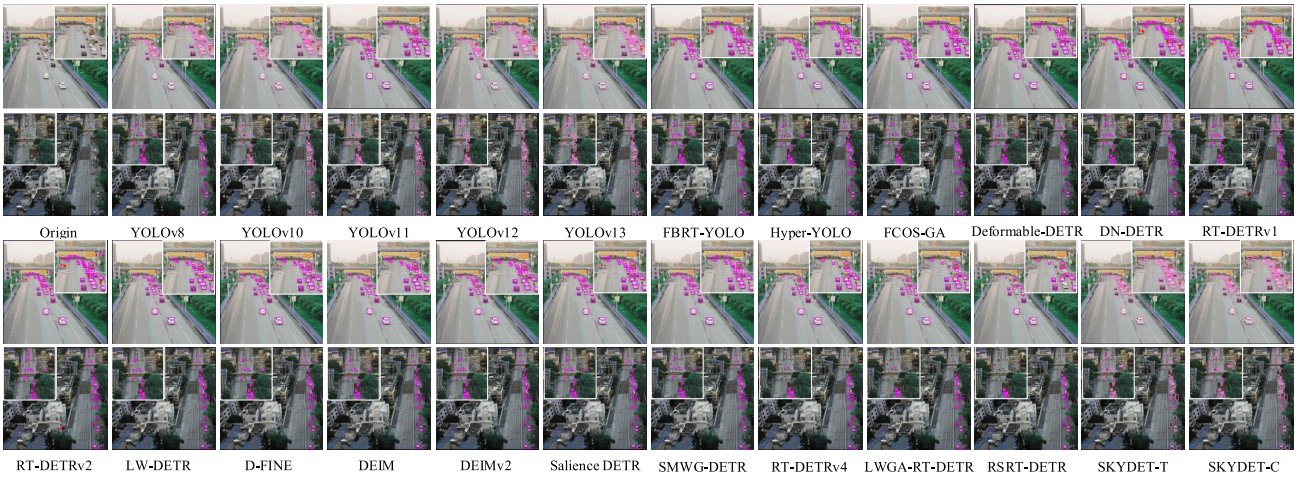


Fig. 6. Display of detection results of different models on the AI-TOD dataset.



Fig. 7. Visualization of the detection performance of SKYDET-T and SKYDET-C on the NWPU VHR-10 dataset.

Furthermore, to validate robustness under extreme scale variations, we also conduct experiments on AI-TOD and NWPU VHR-10. Across these two datasets, which emphasize different viewing perspectives and spatial resolutions, SKYDET consistently maintains SOTA performance. In particular, on AI-TOD, which contains a large number of tiny objects, existing detectors often suffer a sharp drop in AP_{50} due to feature vanishing, whereas SKYDET still achieves 56.1%, exceeding the highly competitive new runner-up RT-DETRv4 by 0.8%. Fig. 6 visualizes partial detection results of two variants of SKYDET and other models on the AI-TOD dataset. This result strongly demonstrates the powerful feature aggregation capability of RGM, enabling accurate localization even when target pixels occupy an extremely small proportion of the image. Meanwhile, on NWPU VHR-10, the superior

TABLE V
ABLATION STUDY OF KEY COMPONENTS IN SKYDET

Group	DINOv3	SGA	CFE	Performance		
				AP_{50}	AP_{75}	$AP_{50:95}$
1	×	×	×	0.645	0.395	0.385
2	✓	×	×	0.668	0.412	0.405
3	✓	✓	×	0.695	0.462	0.450
4	✓	✓	✓	**0.708**	**0.475**	**0.462**

performance of SKYDET on the high-threshold metric AP_{75} further confirms its advantage in fine-grained bounding-box regression. Fig. 7 shows qualitative detection results of two variants of SKYDET on the NWPU VHR-10 dataset.

Finally, considering model parameter size, computational complexity, and detection accuracy jointly, SKYDET exhibits strong competitiveness in the speed–accuracy tradeoff. Although introducing a VFM backbone increases the computational burden of feature extraction to some extent, the efficient architectural design of SKYDET improves detection accuracy without a noticeable increase in the number of parameters or GFLOPs. Compared with lightweight models that prioritize extreme speed at the cost of accuracy, as well as oversized models that pursue maximum accuracy but are difficult to deploy, SKYDET provides a more balanced and efficient solution, demonstrating its strong potential for remote sensing object detection.

Notably, SKYDET's competitive performance on the AI-TOD dataset is no coincidence; rather, it stems from an architectural design specifically tailored for tiny objects. Tiny objects in aerial imagery typically suffer from severe feature degradation and are easily overwhelmed by complex background clutter. SKYDET overcomes this bottleneck through three highly coupled mechanisms. First, by directly transferring the representation space of DINOv3, SKYDET leverages robust, scale-invariant semantic priors to ensure that the faint signals of tiny targets are not obscured by background noise

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TABLE VI
ABLATION STUDY OF COMPONENT EFFECTIVENESS FOR SKYDET-C

Group	DINOv3	CFE	Performance		
			AP_{50}	AP_{75}	$AP_{50:95}$
1	×	×	0.691	0.419	0.410
2	✓	×	0.717	0.430	0.420
3	×	✓	0.706	0.445	0.423
4	✓	✓	**0.726**	**0.471**	**0.450**

during backbone encoding. Second, the proposed SGA module acts as a precise semantic filter; it utilizes the deepest abstract representations from the VFM to generate spatial masks, effectively suppressing irrelevant background clutter and isolating tiny objects before the fusion stage. Finally, the RGM module within the CFE enhances boundary-sensitive responses of tiny objects by using high-level semantics to guide spatial attention and low-level fine-grained details to guide channel attention, thereby effectively mitigating the spatial misalignment inherent in tiny object detection. This mechanism directly drives significant performance gains in high-precision regression metrics.

F. Ablation Study

To verify the effectiveness of each key component in SKYDET and its contribution to overall performance, we conduct ablation studies on the DOTA-v1.0 dataset. For SKYDET-T, we use as the baseline (Group 1) a model that does not use DINOv3–Vision Transformer pretrained weights, does not use SGA, and does not use CFE. Notably, to construct a highly competitive control group, we adopt HGNet v2 [15] as the backbone of the baseline. HGNet v2 is one of the most advanced CNN architectures in real-time detection, known for its strong local feature extraction capability and inference efficiency. Using HGNet v2 with a standard feature pyramid as the baseline can represent the upper bound of mainstream DETR performance, thereby providing a stringent reference for validating the effectiveness of VFMs. For SKYDET-C, we use as the baseline (Group 1) a model that does not use DINOv3–ConvNeXt pretrained weights and does not use CFE. All experiments are conducted under identical training configurations and hyperparameter settings to ensure a fair comparison. The ablation experiment results are shown in Tables V and VI.

From Table V, we observe that replacing the baseline backbone from **HGNet v2** to DINOv3 (Group 2) improves AP_{50} by 2.3%, validating the robustness of features learned via large-scale pretraining. More importantly, introducing SGA to replace the standard FPN (Group 3) leads to a sharp increase of 2.7% and 5.0% in AP_{50} and AP_{75}, respectively. This confirms that SGA effectively overcomes the single-scale limitation of the ViT, and that the generated semantic-guided features are substantially higher in quality than those produced by a conventional FPN. Finally, integrating CFE (Group 4) further optimizes feature alignment, achieving the best AP_{50} of 70.8%. These results indicate a strong synergy among DINOv3, SGA, and CFE, establishing the distinctive advantage of SKYDET-T in high-precision localization.

Furthermore, we conduct dedicated ablation studies on the SKYDET-C variant. As shown in Table VI, replacing the backbone with DINOv3 (Group 2) increases AP_{50} by 2.6%, whereas the gain in AP_{75} is only 1.1%. This suggests that, while the strong semantic representations of VFMs significantly enhance object discovery, efficient feature fusion is still required to achieve precise localization. After introducing CFE into the baseline (Group 3), AP_{75} increases substantially by 3.6%, strongly confirming that RGM effectively addresses feature misalignment and enables high-precision regression by sharpening target boundaries. When DINOv3 and CFE are jointly integrated, the model reaches the best performance, indicating a complementary interplay between robust semantic features provided by the VFM and the fine-grained fusion mechanism of CFE: the former addresses the ability to "detect," while the latter ensures "accurate localization." Their combined effect enables SKYDET to achieve the best results across all metrics, significantly outperforming any variant with a single modification.

TABLE VII
STATISTICAL SIGNIFICANCE AND STABILITY ANALYSIS ACROSS 5 RANDOM SEEDS

Training Strategy	Seed 42	Seed 1024	Seed 2048	Seed 3407	Seed 4096	Mean ± Std
Full Fine-Tuning	0.718	0.714	0.725	0.716	0.724	0.719 ± 0.005
Frozen Backbone	**0.726**	**0.725**	**0.727**	**0.724**	**0.727**	**0.726 ± 0.001**

G. Robustness and Statistical Analysis

To rigorously validate our architecture choice and the freezing strategy for the visual base model, we performed fine-grained statistical evaluations and complexity analyses. Unless otherwise stated, all supplementary experiments in this section were conducted on the SKYDET-C variant on the DOTA-v1.0 dataset. The hardware environment and optimization hyperparameters strictly followed the standard configuration detailed in Section IV-D.

1) Statistical Significance and Stability: We evaluated the SKYDET-C model under both frozen and full-scale fine-tuning settings at five independent random seeds. As shown in Table VII, the freezing strategy not only yielded consistently higher mean AP_{50}, but also a significantly reduced variance (±0.13% versus ±0.48%). Completely unfreezing the large parameter space of DINOv3 on a small-scale aerial dataset leads to highly nonconvex optimized terrain, resulting in catastrophic forgetting and degradation of the representational manifold.

2) Fine-Grained Freezing Ratio: Furthermore, we conducted ablation experiments on the freezing depth of the ConvNeXt backbone network, as detailed in Table VIII. The results show that partial fine-tuning creates a representation mismatch between frozen and fine-tuned VFM layers, disrupting the internally coupled feature hierarchy. Retaining all VFM knowledge (fully frozen) and completely delegating the

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TABLE VIII
FINE-GRAINED ABLATION ON DINOV3-CONVNEXT FREEZING RATIOS ON DOTA-V1.0

Freezing Depth	Tuned Stages	AP_{50}	AP_{75}
Unfreeze All	Stem, Stage 1, 2, 3, 4	0.720	0.449
Freeze Stem + Stage 1	Stage 2, 3, 4	0.722	0.454
Freeze up to Stage 2	Stage 3, 4	0.723	0.461
Freeze up to Stage 3	Stage 4	0.724	0.465
Freeze All (Ours)	**None (Only Adapter)**	**0.726**	**0.471**

domain adaptation task to our CFE proves to be the optimal strategy.

H. Discussion on Experimental Fairness and VFM Capabilities

To provide a more rigorous assessment of SKYDET, it is essential to discuss the disparity in pretraining resources between the DINOv3-based backbone and traditional supervised baselines. While the large-scale pretraining of DINOv3 provides a significant representational advantage, our experiments demonstrate that the architectural innovation of SKYDET is the key factor in successfully transferring this capability to the remote sensing domain.

As shown in the decoupling analysis in Table VI, a raw DINOv3-ConvNeXt backbone without our proposed modules achieves 71.7% AP_{50} on DOTA-v1.0. Although this is higher than supervised baselines like D-FINE (68.9%), it is our CFE that pushes the performance to the ultimate 72.6%. This 0.9% absolute gain under identical pretraining conditions confirms that our reciprocal guidance mechanism effectively addresses the semantic ambiguity and feature misalignment inherent in aerial imagery. Furthermore, compared to DEIMv2, which also utilizes DINOv3 but relies on the ViT architecture, our SKYDET-T (70.8%) still maintains a clear advantage (versus DEIMv2's 70.1%). This establishes that while foundation models provide a strong starting point, dedicated structural adaptation is indispensable for achieving high-precision real-time detection in complex remote sensing scenes.

Furthermore, setting aside the performance gains derived from DINOv3 pretraining, we conducted strictly controlled cross-architecture comparisons by seamlessly integrating our CFE into other mainstream detectors. The consistent AP improvements observed across various baseline models clearly demonstrate that our reciprocal guidance mechanism functions as a highly generalizable, plug-and-play component capable of effectively mitigating feature alignment issues—an advantage that exists independently of the base model's inherent representational strengths.

I. Visualization and Qualitative Analysis

To qualitatively analyze the detection capability of SKYDET, we employ gradient-weighted class activation mapping (Grad-CAM) [60] to visualize heatmaps of SKYDET and other object detectors on the DOTA-v1.0 dataset. As shown in Fig. 8, compared with other models, SKYDET exhibits a noticeably more "focused" activation pattern: high-response regions (red) precisely cover the target objects, while responses in background areas are effectively suppressed (blue). This high-contrast attention distribution demonstrates that RGM successfully leverages high-level semantics to filter low-level details, enabling the model to identify the true regions of interest from complex backgrounds.

J. Comparative Analysis of DINOv3 Architectures

The architectural design of a VFM determines the granularity and semantic strength of its feature representations. To identify the most suitable feature extractor, we conduct a detailed evaluation of different DINOv3 pretrained weights on the DOTA-v1.0 dataset. Specifically, we systematically evaluate the ViT and ConvNeXt architectures across various scales under both full fine-tuning and frozen backbone strategies. The results are reported in Table IX.

As shown in Table IX, for ViT backbones, because the network can only output single-scale features, smaller pretrained weights remain constrained by the inherent limitation of single-scale representations even after fine-tuning. Without our proposed feature fusion network, ViT-S/16 achieves only 24.9% AP_{50} even under the full fine-tuning strategy. This highlights that the standard ViT architecture, without a hierarchical feature pyramid, struggles to capture the high-frequency spatial details that are critical for detection in remote sensing imagery. Meanwhile, the results also strongly validate the effectiveness of our feature adapter: after introducing the fusion module, the ViT family achieves a substantial performance leap. Under the frozen setting, ViT-S/16 improves to 63.7% AP_{50}, and ViT-H+/16 further reaches 69.5%, indicating that SGA successfully reconstructs ViT's single-scale semantic features into more adaptable multiscale spatial representations, making it more suitable for remote sensing object detection. In addition, although the ViT is not particularly strong in terms of AP_{50}, it exhibits an advantage in high-precision localization, while its sensitivity to small objects is comparable to ConvNeXt. As shown in the table, after incorporating SGA, larger ViT models significantly surpass ConvNeXt on $AP_{50:95}$ regardless of whether full fine-tuning is used, while achieving comparable performance on AP_S. For applications that prioritize extreme localization accuracy and have sufficient computational resources, the ViT is therefore a preferable choice.

In contrast, the ConvNeXt architecture shows stronger adaptability in the tradeoff between computational efficiency and detection recall. Although it is slightly inferior to the ViT on the high-precision metric $AP_{50:95}$, ConvNeXt performs notably well on AP_{50}, which emphasizes object discovery. With the feature aggregation network, ConvNeXt-Large reaches 71.0% AP_{50} under the frozen setting and further achieves 72.8% with full fine-tuning, clearly outperforming all ViT scales and demonstrating competitive detection performance. This efficient feature representation can be largely attributed to the inductive bias inherent to CNNs, which enables a more compact architecture to capture key spatial details when handling dense small objects with drastic scale variations in remote sensing imagery. Rather than concluding

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TABLE IX

QUANTITATIVE RESULTS OF THE BACKBONE USING DIFFERENT PRETRAINED WEIGHTS AND BASIC FINE-TUNING STRATEGIES IN THE DOTA V1.0 DATASET ARE SHOWN. TRUE AND FALSE INDICATE WHETHER THE NETWORK USES THE NECK NETWORK FOR FEATURE AGGREGATION. FULL MEANS THAT THE ENTIRE NETWORK IS FINE-TUNED USING PRETRAINED WEIGHTS, AND FREEZE MEANS THAT THE PARAMETERS IN THE BACKBONE NETWORK ARE FROZEN AND ONLY THE SUBSEQUENT FEATURE AGGREGATION AND DETECTION PARTS ARE TRAINED. TRAINABLE PARAMS REPRESENT THE NUMBER OF TRAINABLE PARAMETERS IN THE BACKBONE (EXCLUDING THE SUBSEQUENT FEATURE AGGREGATION AND DETECTION PARTS)

Pretrained Backbone	Fusion	Weights/Scale	Strategy	Params (M)	AP_S	AP_M	AP_L	AP_{50}	AP_{75}	$AP_{50:95}$
ViT Based on LVD-1689M Dataset	False	ViT-S/16	Full	16.53	0.052	0.132	0.169	0.249	0.094	0.115
			Freeze	0	0.012	0.041	0.050	0.080	0.016	0.030
		ViT-S+/16	Full	16.53	0.030	0.079	0.108	0.158	0.053	0.070
			Freeze	0	0.011	0.031	0.043	0.075	0.016	0.028
		ViT-B/16	Full	112.4	0.107	0.241	0.317	0.446	0.201	0.226
			Freeze	0	0.026	0.073	0.087	0.146	0.042	0.060
		ViT-L/16	Full	298.2	0.132	0.290	0.389	0.526	0.261	0.278
			Freeze	0	0.021	0.057	0.080	0.127	0.033	0.050
		ViT-H+/16	Full	632.0	0.099	0.236	0.333	0.447	0.207	0.228
			Freeze	0	0.008	0.025	0.027	0.056	0.010	0.020
		ViT-7B/16	Full	6900	0.085	0.205	0.290	0.410	0.175	0.195
			Freeze	0	0.005	0.012	0.018	0.035	0.006	0.015
	True	ViT-S/16	Full	40.21	0.251	0.411	0.494	0.645	0.395	0.398
			Freeze	0	0.237	0.409	0.484	0.637	0.392	0.392
		ViT-S+/16	Full	85.88	0.249	0.417	0.498	0.645	0.404	0.401
			Freeze	0	0.243	0.416	0.503	0.650	0.395	0.401
		ViT-B/16	Full	171.5	0.272	0.455	0.535	0.680	0.445	0.435
			Freeze	0	0.253	0.432	0.494	0.659	0.417	0.409
		ViT-L/16	Full	458.2	0.288	0.475	0.565	0.695	0.465	0.452
			Freeze	0	0.255	0.443	0.508	0.673	0.421	0.420
		ViT-H+/16	Full	805.5	0.295	0.488	0.580	0.705	0.475	0.462
			Freeze	0	0.282	0.472	0.565	0.695	0.462	0.450
		ViT-7B/16	Full	7200	0.302	0.495	0.595	0.712	0.482	0.468
			Freeze	0	0.290	0.485	0.580	0.702	0.470	0.458
ViT Based on SAT-493M Dataset	False	ViT-L/16	Full	298.2	0.111	0.254	0.359	0.481	0.228	0.251
			Freeze	0	0.021	0.068	0.080	0.133	0.035	0.053
		ViT-7B/16	Full	6900	0.075	0.180	0.255	0.380	0.155	0.175
			Freeze	0	0.005	0.010	0.015	0.030	0.005	0.012
	True	ViT-L/16	Full	458.2	0.282	0.468	0.555	0.688	0.460	0.448
			Freeze	0	0.258	0.436	0.507	0.669	0.418	0.417
		ViT-7B/16	Full	7200	0.298	0.488	0.585	0.705	0.475	0.463
			Freeze	0	0.285	0.478	0.570	0.695	0.465	0.453
ConvNeXt Based on LVD-1689M Dataset	False	Tiny	Full	23.3	0.262	0.424	0.496	0.695	0.410	0.402
			Freeze	0	0.190	0.344	0.415	0.601	0.307	0.325
		Small	Full	44.9	0.272	0.436	0.507	0.712	0.413	0.413
			Freeze	0	0.191	0.355	0.425	0.616	0.324	0.335
		Base	Full	83.1	0.275	0.435	0.502	0.708	0.407	0.406
			Freeze	0	0.195	0.362	0.434	0.625	0.208	0.344
		Large	Full	192.1	0.299	0.449	0.528	0.718	0.438	0.426
			Freeze	0	0.207	0.369	0.455	0.635	0.338	0.351
	True	Tiny	Full	31.3	0.269	0.434	0.503	0.708	0.421	0.411
			Freeze	0	0.276	0.435	0.494	0.695	0.406	0.403
		Small	Full	53.0	0.280	0.447	0.514	0.717	0.430	0.420
			Freeze	0	0.274	0.437	0.508	0.699	0.418	0.411
		Base	Full	91.2	0.295	0.444	0.520	0.720	0.428	0.421
			Freeze	0	0.285	0.447	0.503	0.700	0.430	0.414
		Large	Full	200.1	0.301	0.452	0.520	0.728	0.436	0.426
			Freeze	0	0.288	0.453	0.508	0.710	0.429	0.418

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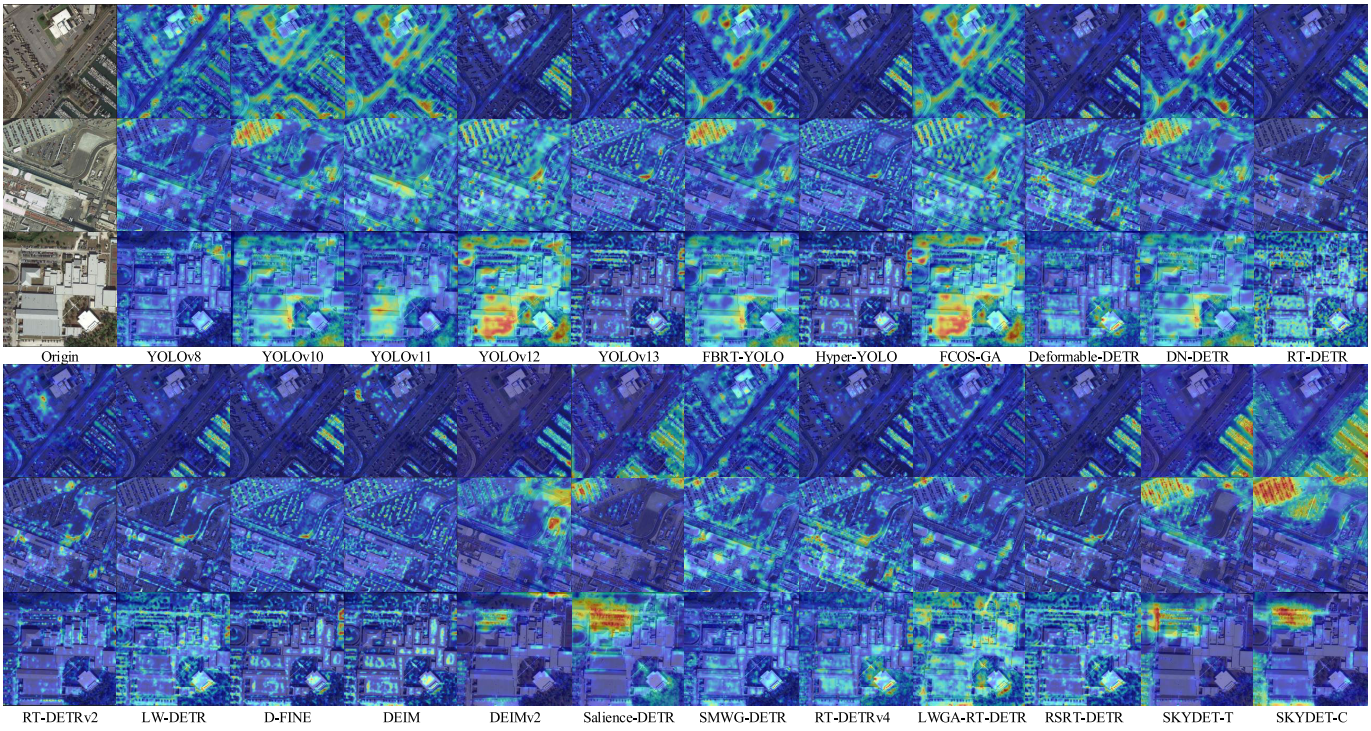


Fig. 8. Visualization of SKYDET and other detection models under Grad-CAM: the brighter the image, the higher the model's attention to that region.

that one architecture strictly supersedes the other, our analysis highlights their complementary nature. It is crucial to emphasize that the high-precision localization capability of the ViT backbone would be entirely inaccessible in aerial scenes without our proposed SGA, which successfully salvages the ViT from severe feature degradation (boosting AP_{50} from a mere 24.9%–63.7% under the frozen setting).

Considering both detection speed and accuracy, we finally propose two parallel and complementary SKYDET variants: SKYDET-C with a ConvNeXt backbone, which is more robust for high-recall object discovery and practical deployment, and SKYDET-T with a ViT backbone enabled by SGA, which is highly recommended for tasks demanding stringent bounding-box regression. Specifically, SKYDET-T adopts the frozen ViT-H+/16 distilled model with SGA, while SKYDET-C adopts a frozen Base model. Furthermore, practical deployment tests on a single NVIDIA A10 GPU (TensorRT, FP16) demonstrate distinct deployment capabilities for our variants. SKYDET-C achieves a highly competitive real-time inference speed of 52 frames/s, successfully maintaining an optimal speed-accuracy tradeoff for real-time aerial image analysis. In contrast, SKYDET-T operates at approximately 8 frames/s, exchanging inference speed for extreme localization precision, making it more suitable for rigorous offline mapping applications.

K. Impact of Pretraining Paradigms and Backbone Selection

To investigate the deeper impact of pretraining paradigms and network architectures on remote sensing detection performance, and to verify that DINOv3 remains highly competitive for remote sensing object detection, we conduct an extensive comparative study. Specifically, we integrate multiple representative architectures—including CNNs, ViT, and state-space models (SSMs)—into D-FINE (L), while keeping the feature fusion network and detection head identical. We ensure that all backbones load their officially released best pretrained weights, and all other settings follow those in Section IV-J. Notably, for the choice of DINOv3 model scale, after jointly considering detection speed and accuracy, we adopt the configurations described in Section IV-J, where DINOv3-T denotes the ViT-H+/16 distilled model equipped with SGA, and DINOv3-C denotes the ConvNeXt-Base model. The ablation experiment results are shown in Table X.

From the results, we observe that traditional supervised models such as VGG16 and HGNetv2 suffer significant performance degradation under the frozen strategy, with AP_{50} far below their full fine-tuning counterparts. This phenomenon is mainly attributed to the large domain gap between ImageNet natural scenes and the aerial top-down perspective of remote sensing imagery, as well as the task-specific nature of supervised features, making the frozen backbone unable to extract effective discriminative representations from previously unseen aerial images. Furthermore, when comparing representative VFMs, different pretraining paradigms exhibit distinct impacts. MAE yields suboptimal performance under the frozen setting, as pure pixel-level reconstruction lacks the object-centric semantic priors essential for isolating tiny aerial targets. While CLIP, EVA-02, and DINOv2 exhibit highly competitive results with different large-scale pretraining objectives, they still fall short of DINOv3. Ultimately, benefiting from pretraining on 1.7 billion images and its advanced self-distillation paradigm, DINOv3 exhibits remarkable robustness in the frozen setting. In particular, DINOv3-C achieves 70.0% AP_{50} even with frozen parameters, which not only substantially surpasses

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TABLE X
COMPARISON OF DETECTION PERFORMANCE WITH DIFFERENT PRETRAINED BACKBONES UNDER FULL FINE-TUNING AND FREEZE STRATEGIES

Backbone	Strategy	Params (M)	AP_S	AP_M	AP_L	AP_{50}	AP_{75}	$AP_{50:95}$
VGG16 [61]	Full	138.3	0.165	0.352	0.410	0.602	0.354	0.331
	Freeze	0	0.052	0.124	0.158	0.254	0.121	0.105
HGNetv2-H [51]	Full	54.0	0.265	0.440	0.515	0.689	0.402	0.398
	Freeze	0	0.155	0.295	0.330	0.468	0.290	0.265
ConvNeXt-L [62]	Full	197.8	0.291	0.448	0.522	0.718	0.438	0.426
	Freeze	0	0.205	0.375	0.460	0.635	0.338	0.351
VMamba-B [63]	Full	89.0	0.288	0.465	0.535	0.712	0.430	0.418
	Freeze	0	0.180	0.350	0.410	0.525	0.315	0.305
HGNet v2 [15]	Full	304.5	0.275	0.455	0.540	0.692	0.445	0.435
	Freeze	0	0.125	0.280	0.350	0.485	0.255	0.240
MAE-Base [64]	Full	86.0	0.260	0.430	0.520	0.698	0.405	0.412
	Freeze	0	0.130	0.290	0.360	0.490	0.260	0.245
EVA-02-Base [65]	Full	86.0	0.285	0.460	0.540	0.714	0.436	0.445
	Freeze	0	0.240	0.410	0.480	0.650	0.395	0.380
CLIP [24]	Full	304.3	0.250	0.425	0.510	0.665	0.425	0.410
	Freeze	0	0.230	0.390	0.450	0.615	0.385	0.365
DINOv2 [21]	Full	304.4	0.288	0.475	0.565	0.695	0.465	0.452
	Freeze	0	0.255	0.443	0.508	0.673	0.421	0.420
DINOv3-T	Full	805.50	0.295	0.488	0.580	0.705	0.475	0.462
	Freeze	0	0.282	0.472	0.565	0.695	0.462	0.450
DINOv3-C	Full	91.17	0.295	0.444	0.520	0.720	0.428	0.421
	Freeze	0	0.285	0.447	0.503	0.700	0.430	0.414

supervised models and other VFMs under the same strategy, but also exceeds the fully fine-tuned SOTA architecture VMamba-B. These results strongly suggest that, when tackling complex background interference and domain shift in remote sensing imagery, the scale and quality of pretraining data can be more critical than sophisticated architectural design, and that discriminative self-supervised learning can endow models with stronger general-purpose visual representations.

L. Reciprocal Guidance Ablation

To validate the effectiveness of the "reciprocal guidance between spatial structure and channel semantics" strategy in RGM, as well as the rationality of the guidance direction, we conduct structured ablation experiments on the DOTA-v1.0 dataset. Except for the replaced feature fusion module, all other training configurations are kept identical. We construct the following six variants for comparison.

1) *Baseline (Concat):* As the baseline, we simply perform feature concatenation followed by a 1×1 convolution for cross-level fusion, without any attention-guided mechanism.
2) *Single-S (Uni-Directional Spatial):* Only retains uni-directional spatial guidance, that is, generating a spatial attention map M_s from high-level features to enhance low-level features, while ignoring reverse guidance along the channel dimension.
3) *Single-C (Uni-Directional Channel):* Only retains uni-directional channel guidance, that is, generating a channel attention vector M_c from low-level features to reweight high-level features, while ignoring reverse guidance along the spatial dimension.
4) *Swapped (Inverted Guidance):* Swaps the guidance directions, that is, forcing high-level features to generate channel weights and low-level features to generate spatial masks.
5) *Comparison (CBAM):* Replaces RGM with the generic attention module CBAM [66].
6) *Reciprocal (Ours):* Uses the proposed reciprocal guidance strategy, that is, generating spatial saliency from high-level semantics to filter low-level details ($M_s \rightarrow X_l$), and generating channel descriptors from low-level textures to strengthen high-level semantics ($M_c \rightarrow X_h$).

As shown in Table XI, the ablation results and the complexity-versus-gain analysis strongly support the effectiveness and efficiency of RGM for remote sensing object detection. First, compared with the baseline, both Single-S and -C achieve noticeable improvements, confirming the necessity of introducing attention mechanisms into cross-scale feature fusion. However, their performance remains clearly inferior to the proposed Reciprocal strategy, indicating that uni-directional guidance along a single dimension cannot fully exploit the complementarity of multiscale features. Second, the guidance direction is critical. The results show that the

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TABLE XI

ABLATION STUDY OF DIFFERENT GUIDANCE STRATEGIES AND THEIR COMPLEXITY-VERSUS-GAIN ANALYSIS. THE PROPOSED RECIPROCAL STRATEGY ACHIEVES THE BEST PERFORMANCE WITH NEGLIGIBLE COMPUTATIONAL OVERHEAD

Method	Guidance Strategy	AP_{50}	AP_{75}	$AP_{50:95}$	AP_S	Params (M)	GFLOPs
Baseline	Concat	0.712	0.436	0.425	0.283	90.85	236.1
Single-S	High → Low	0.718	0.452	0.434	0.309	91.02	236.9
Single-C	Low → High	0.720	0.449	0.436	0.298	90.98	236.5
Swapped	Inverted Direction	0.713	0.437	0.426	0.285	91.17	237.8
Comparison	Attention (CBAM [66])	0.722	0.460	0.442	0.315	91.10	237.2
Reciprocal	**Reciprocal**	**0.726**	**0.471**	**0.450**	**0.331**	**91.17**	**237.8**

Swapped variant leads to a significant performance drop, even underperforming the uni-directional settings on some metrics. This suggests that high-level features in remote sensing imagery, which contain rich semantic information, are more suitable for generating spatial masks to localize targets, whereas low-level features, which preserve abundant textures and edge details, are better suited for producing channel weights to select the most discriminative feature maps. By leveraging this complementarity, RGM achieves the best overall performance. Furthermore, the complexity evaluation reveals that this significant performance leap (+1.4% AP_{50} and +3.5% AP_{75} over the baseline) is achieved at a highly cost-effective margin, adding merely 0.32M parameters and 1.7 GFLOPs. Notably, the gains brought by RGM are particularly pronounced on high-threshold metrics such as AP_{75} and $AP_{50:95}$. This indicates that, by alleviating feature misalignment and semantic ambiguity during cross-scale fusion, RGM substantially improves bounding-box regression accuracy and localization quality without incurring a heavy computational burden, making it especially suitable for remote sensing observation scenarios with complex backgrounds and small objects.

V. CONCLUSION

To address the scale mismatch and feature adaptation challenges of general-purpose large vision models in complex remote sensing scenarios, we propose SKYDET, a high-performance object detection framework for remote sensing imagery. We first conduct a systematic evaluation and in-depth analysis of different architectural variants of DINOv3, demonstrating that it still possesses strong feature extraction capability when transferred to remote sensing object detection. We further design a novel multiscale feature fusion adapter, termed SGA, to overcome the limitations of the ViT architecture in fine-grained object detection. This module effectively reconstructs deep semantic representations from the foundation model into spatially sensitive hierarchical features, substantially enhancing the sensitivity of ViTs to tiny objects. Although the ViT can reconstruct multiscale features with the proposed SGA, we find that the ConvNeXt architecture, benefiting from its inherent inductive bias and hierarchical structure, exhibits superior adaptability in capturing fine-grained spatial information in remote sensing imagery; thus, we ultimately propose SKYDET-C as our primary robust detector for general scenarios, alongside SKYDET-T for scenarios requiring extreme localization accuracy. Finally, to break the common bottlenecks of semantic ambiguity and feature misalignment in cross-scale fusion, we introduce a CFE, whose core module RGM establishes a reciprocal enhancement mechanism that mutually calibrates high-level semantic localization and low-level texture details, effectively suppressing background noise and significantly sharpening boundary responses of dense small objects. Experimental results on DOTA-v1.0, NWPU VHR-10, and AI-TOD demonstrate that SKYDET achieves outstanding effectiveness and competitive accuracy, strongly validating the effectiveness of transferring large-scale pretrained knowledge to remote sensing tasks and highlighting the crucial role of model scale in compensating for data scarcity.

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